

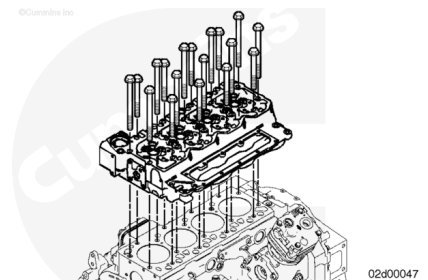
Trainings ▾

Install

B4.5 RGT Engines

⚠ WARNING ⚠

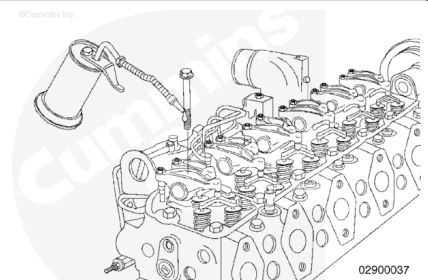
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Carefully put the cylinder head on the cylinder block and seat it onto the dowels.

Lubricate the threads and under the heads on the cylinder head mounting capscrews with clean engine oil.

Install the capscrews and tighten finger-tight.



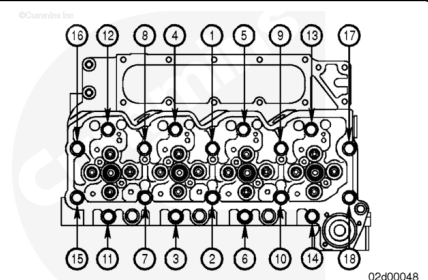
Use the illustrated sequence to tighten the cylinder head capscrews.

Tighten the capscrews.

Torque Value:

All Capscrews

1. 90 n•m [66 ft-lb]
2. 90 n•m [66 ft-lb]
3. Rotate 90-degrees clockwise.

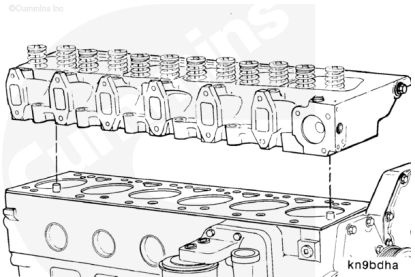


B3.9, B5.9, and B4.5 Engines

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Carefully put the cylinder head straight down onto the cylinder block and seat it onto the dowels.

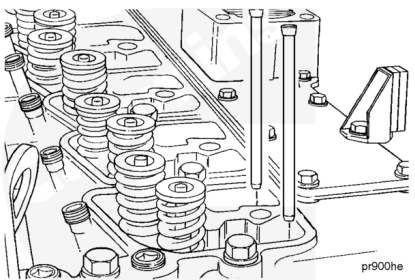


Cylinder Head Weight

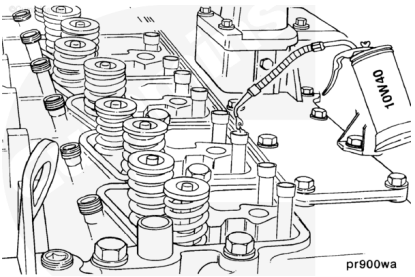
Cylinder Number	Kg	lb
4	36	79
6	51.3	113

Push Tubes - Installation

Position the push tubes into the valve tappets.

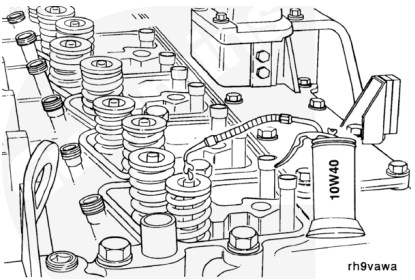


Lubricate the push tube sockets with clean lubricating engine oil.

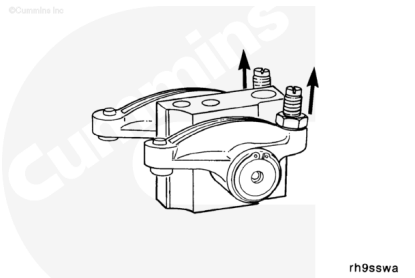


Rocker Levers - Installation

Lubricate the valve stems with clean lubricating engine oil.

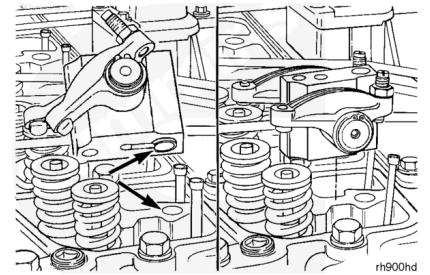


Completely loosen the rocker lever adjusting screws.



Note : The rocker lever pedestals are aligned with dowels.

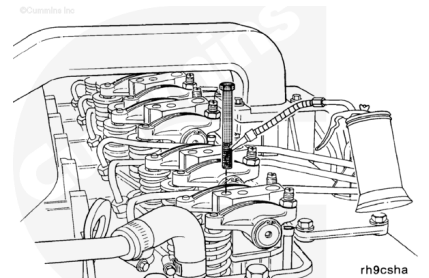
Install the pedestals.



Use capscrew length gauge, Part Number 3823921, to inspect all cylinder head capscrews for proper length.

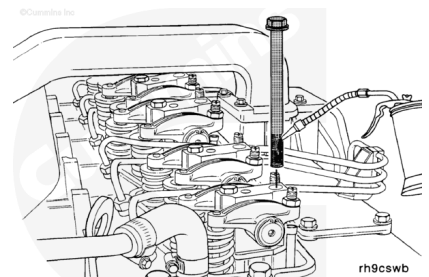
Lubricate the 8-mm pedestal capscrew threads and under the capscrew heads with clean lubricating engine oil.

Install the capscrews finger-tight.



Lubricate the 12-mm pedestal/head capscrew bolt threads and under the capscrew heads with clean lubricating engine oil.

Install the capscrews finger tight.

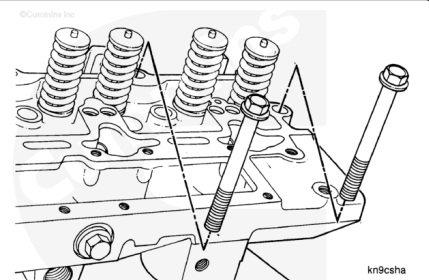


Lubricate the threads and under the heads on the remaining cylinder head capscrews with clean lubricating engine oil.

Install capscrews in the cylinder head and finger-tighten.

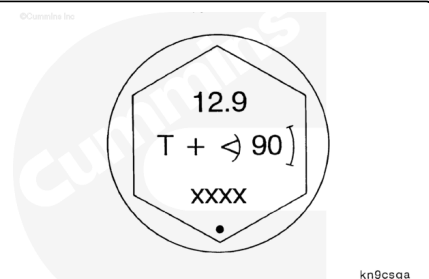
Note : Be sure to install the long capscrews into the holes underneath the injectors.

Note : Capscrews for the 1991 and later certification level engines are the same overall length, but have a longer threaded area.



⚠ CAUTION ⚠

Do not use pre-1991 certification level engine capscrews in a 1991 B Series or later certification level engine because pre-1991 capscrews do not have enough threads to provide enough thread engagement in the 1991 and later certification engines. Failure to do so can result in engine damage.



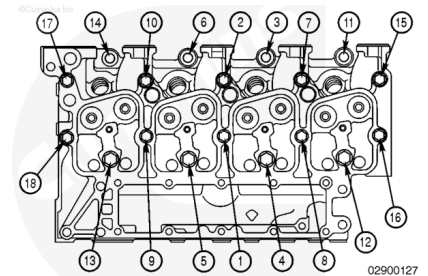
⚠ CAUTION ⚠

Capscrews for a 1991 B Series or later certification level engine can be used in a pre-1991 certification level engine because 1991 and later capscrews have enough threads to provide enough engagement in all certification level engines.

Note : The top of the cylinder head capscrew is identified with an angle marking. The cylinder head capscrews **must** be tightened by using the three-step “torque plus angle” method described as follows.

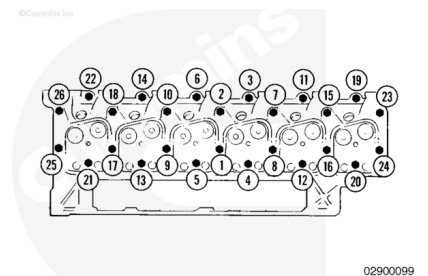
On a four cylinder engine, capscrew number 1 is located in between cylinders 2 and 3. The numbered sequence is the same as a six cylinder, but stops at capscrew number 18. Follow the numbered sequence for the four cylinder engine, and tighten all 18 capscrews.

Torque Value: 90 n•m [66 ft-lb]



Follow the numbered sequence for the six cylinder engine, and tighten all 26 capscrews.

Torque Value: 90 n•m [66 ft-lb]



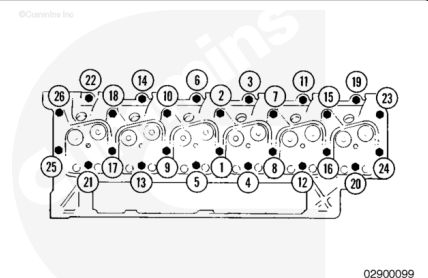
Four Cylinder

Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4,5,12, and 13).

Six Cylinder

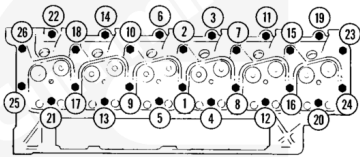
Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4, 5,12, and 13, 20, and 21).

Torque Value: 120 n•m [89 ft-lb]



Tighten the short capscrews again (numbers 1, 2, 3, 6, 7, 8, 9, 10, 11, 14, 15, 16, 17, 18, 19, 22, 23, 24, 25, and 26) because of cylinder head relaxation and to obtain proper cylinder head torque requirements.

Torque Value: 90 n•m [66 ft-lb]



02900099

Tighten the long capscrews again because of cylinder head relaxation and to obtain proper cylinder head torque requirements.

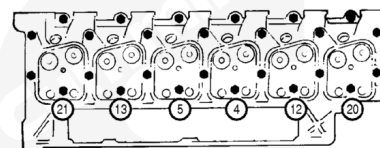
Four Cylinder

Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4, 5, 12, and 13).

Six Cylinder

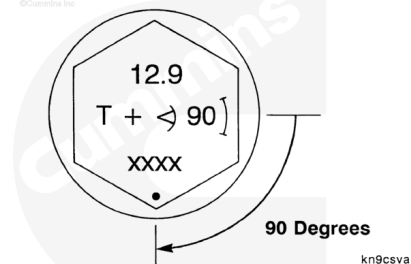
Follow the numbered sequence, and tighten the long capscrews **only** (numbers 4, 5, 12, and 13, 20, and 21).

Torque Value: 120 n·m [89 ft-lb]



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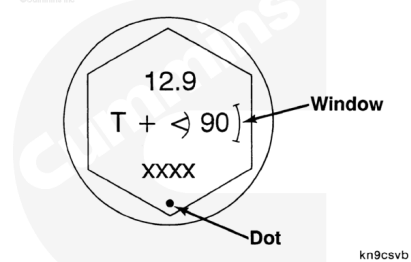
Follow the numbered sequence, and turn the capscrew 90 degrees as indicated on the capscrew head.



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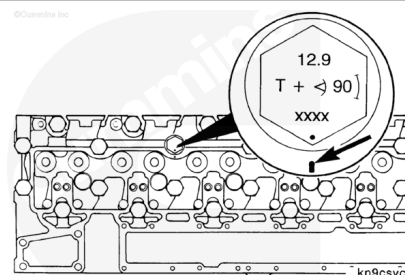
Torque Angle Gauge - 3/4-Inch Drive, Tool Number 3823878

To turn the capscrew to the desired angle accurately, align the capscrew with the small “dot” and “window” that are marked on the capscrew head or use torque angle gauge - 3/4-inch drive tool, Part Number 3823878.



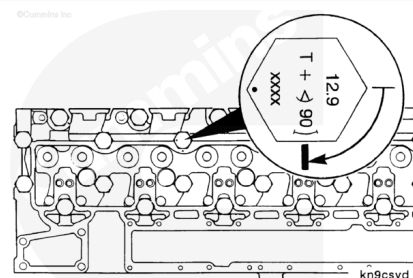
kn9csvgb

Mark the cylinder head adjacent to the dot on the capscrew head. This mark will serve as an indexing aid.



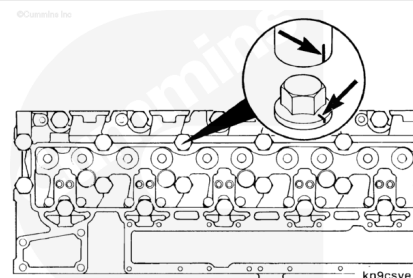
kn9csvgc

Rotate the capscrew until the mark that has been made on the cylinder head falls into the window on the capscrew head.

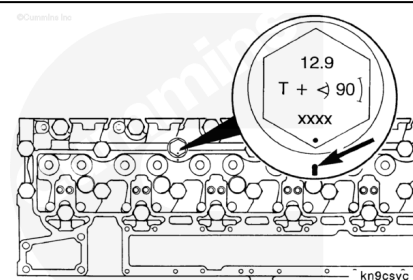


Service Tip:

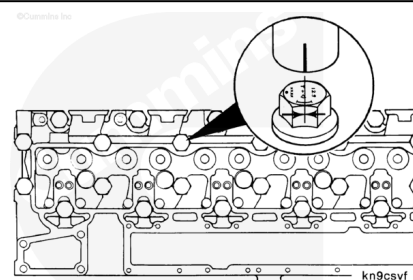
Use a permanent marker to mark the socket corresponding to one of the flats of the socket hex.



After the torque has been applied, mark the cylinder head at the location of the dot.



Position the socket on the capscrew such that the mark on the socket is at the same point as the window on the capscrew.



Turn the socket until the mark on the socket aligns with the mark on the cylinder head.

